Gebze Technical University

CSE 341 — Concepts of Programming Languages · Spring 2026

Project Part 1 — Retrospective (D5)

ScoreScript

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What is Working

The lexer and parser are fully working. All three valid programs parse successfully and produce correct AST output. The --dump-ast flag works as expected. All five malformed programs are rejected with clear error messages and correct line numbers. The grammar design is clean and unambiguous every construct is uniquely identified by its leading token.

What Was Hard

The hardest part was designing the language from scratch. Coming up with the domain and deciding what to include took a lot of thought at the beginning. Writing the EBNF grammar was challenging, especially getting the if/else/for structure right. I also had to keep things simple enough for the parser to handle without getting too complex. Another challenge was making sure the grammar was unambiguous.

What I Plan to Fix in Part 2

In Part 2 I plan to add a type checker and an interpreter. The type checker will catch type errors before the program runs . For example, using a void function result as a value. The interpreter will execute ScoreScript programs end-to-end: average will compute real averages, curve will apply the [0,100] clamp, rank will sort students by average, and transcript will print formatted student reports. I also plan to complete the remaining D1 sections , type system, expressions, semantics, and design rationale.